LSGI3245 Geospatial Database Management and Design

2026/27 Semester 1

Final Project Instruction

1. Introduction

In this final project, students will have an opportunity to explore different types of indoor and outdoor sports facilities in Hong Kong, and then be asked to organize some spatial and non-spatial datasets into a relational database. The database created by the students will be used to answer some meaningful questions related to public health.

Research background: Since 2014, Hong Kong has consecutively held the title of the region with the longest life expectancy in the world for ten years. The average life expectancy of Hong Kong residents has risen to the highest globally, reaching 85.29 years. There are numerous factors contributing to the longevity of Hongkongers, such as a well-developed healthcare system, healthy eating habits, favorable environmental and climatic conditions, and a low smoking rate. Although the primary driving forces behind the long-life expectancy of Hong Kong residents still require further investigation, it is widely acknowledged that outdoor activities and sports play a crucial role in the health and well-being of Hong Kong residents.

Upon the completion of this project, students are expected to achieve following **outcomes**:
– Use PostgreSQL and PostGIS to organize some spatial and non-spatial datasets into a relational database.
– Perform some spatial and non-spatial queries over the database to answer meaningful questions about the geographic coverage of health facilities in Hong Kong as well as their relationships with the underlying population distribution (e.g., which areas have a good/poor coverage of indoor and outdoor sports facilities, and what types of facilities are scarce).

2. Data and Study Area

The study area will be Hong Kong, and students will be provided with a collection of files that contain useful information about different types of outdoor and sports facilities in Hong Kong, including 8 different types of data:

– Badminton Courts
– Basketball Courts
– Tennis Courts
– Country Parks
– Fitness Rooms
– Parks, Zoos and Gardens
– Sports Grounds
– Swimming Pools

These files are publicly available and can be found at:
– https://www.csdi.gov.hk/zh-hk

You can search and find the latest dataset on above website like this:

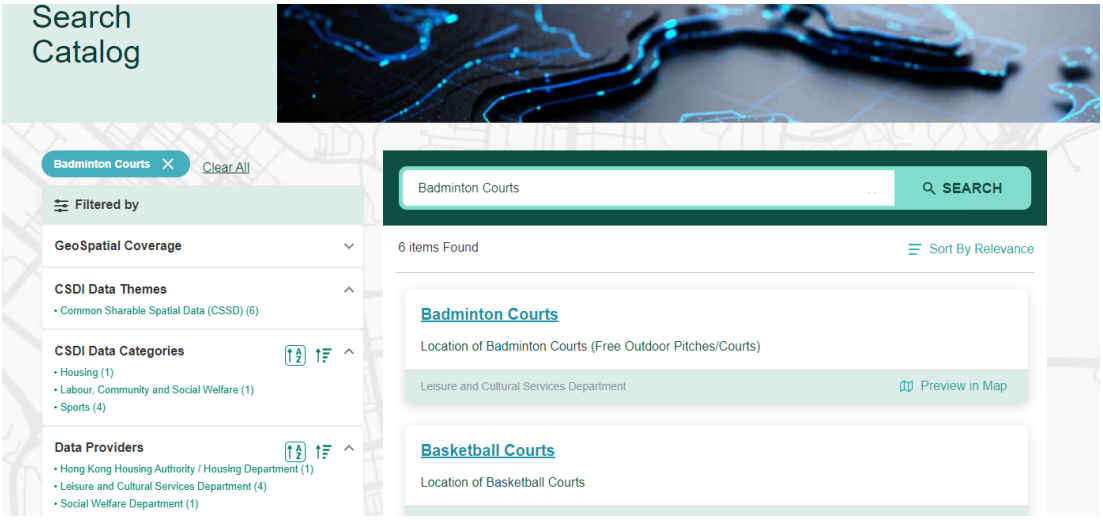


Fig 1. Screenshot of search for dataset 'BadmintonCourts'

Then download the .csv dataset (eg: BadmintonCourts_CSV) and open like this:

A	B	C	D	E	F	G	H	I	J	K
ADDRESS_	ADDRESS_	DATASET_	DATASET_	EASTING	LASTUPDA	LATITUDE	LONGITUD	NAME_EN	NAME_TC	NORTHING
3/F, Tai Po	大埔東昌街	Sports Cer	體育館	835563.2	26/9/2022	22.44989	114.17	Tung Chec	東昌街體育	834493.6
2/F., Sham	九龍深水埗	Sports Cer	體育館	833408.1	15/9/2022	22.33051	114.1491	Sham Shui	深水埗體育	821275
6/F, Tai Po	大埔鄉事會	Sports Cer	體育館	835190.2	22/8/2022	22.44629	114.1664	Tai Po Hui	大埔墟體育	834095.7
52 Lei King	西灣河鯉景	Sports Cer	體育館	840948	8/4/2022 1	22.28473	114.2223	Island East	港島東體育	816205
58 Choi Wi	觀塘彩榮路	Sports Cer	體育館	840352.3	30/11/202	22.32935	114.2165	Choi Wing	彩榮路體育	821146.3
4/F, Cheun	青衣長發邨	Sports Cer	體育館	828627.9	19/2/2021	22.36243	114.1031	Cheung Fa	長發體育館	824815.8
No. 10, Sha	沙田沙田頭	Sports Cer	體育館	837164	6/7/2020 1	22.37223	114.1856	Che Kung	車公廟體育	825894
2-5/F Tuer	新界屯門兆	Sports Cer	體育館	815911	29/6/2020	22.38425	113.9792	Siu Lun Sp	兆麟體育館	827239
29 Cotton	中環紅棉路	Sports Cer	體育館	834456	3/12/2018	22.27734	114.1593	Hong Kon	香港公園	815387
53 Wing Sl	荃灣永順街	Sports Cer	體育館	829651.8	3/12/2018	22.36596	114.1126	Tsuen War	荃灣體育館	825200.7
11/F Sheur	上環皇后大	Sports Cer	體育館	833460	3/12/2018	22.286	114.1496	Sheung W	上環體育館	816346
5/F Shek T	皇后大道西	Sports Cer	體育館	832061	3/12/2018	22.28572	114.1361	Shek Tong	石塘咀體育	816315
4/F, Smithf	西區士美非	Sports Cer	體育館	831297	3/12/2018	22.28194	114.1286	Smithfield	士美非路	815897
18 Eastern	西營盤東邊	Sports Cer	體育館	832828	3/12/2018	22.29017	114.1435	Sun Yat Se	中山紀念公	816808
Unit 202, 2	元朗朗屏邨	Sports Cer	體育館	820449	9/10/2018	22.4503	114.0232	Long Ping	朗屏體育館	834547

Fig 2. Screenshot of data format for 'BadmintonCourts'.

Files for other indoor and outdoor sports facilities have similar data structures. In the final project, students will be asked to clean the data on their own, and then organize these datasets into the same format. Students will also be provided with a Hong Kong map (in the format of shapefile):



Fig 3. Map data for Hong Kong (shown in Google Earth).

Besides that, students will also be provided with the census data that record the number and percentage of the population in Hong Kong at the administrative district level. As shown in Fig 4, and students are recommended to use the latest information collected in 2016 for their final project.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1						按性別劃分的人口		性別比率（即男性數目與每千名女性相對的比率）	按年齡組別劃分的人口					按性別劃分的年齡中位數		
2	區議會分區（分組）	區議會分區	區議會分區（英文名稱）	區議會分區（中文名稱）	總人口	男性	女性		15歲以下	15至24歲	25至44歲	45至64歲	65歲及以上	男女合計	男性	女性
3						Population by sex		Sex ratio (i.e. the number of males per 1 000 females in the respective age group)	Population by age group					Median age by sex		
4	District Council district (class)	District Council district	Name of District Council district (in English)	Name of District Council district (in Chinese)	Total population	Male	Female		< 15	15 - 24	25 - 44	45 - 64	65+	Both sexes	Male	Female
5	dc_class	dc	dc_eng	dc_chi	t_pop	pop_m	pop_f	sr	age_1	age_2	age_3	age_4	age_5	t_ma	ma_m	ma_f
6	A	11	Central and Western	中西區	235 953	98 282	137 671	714	24 382	16 779	77 767	71 534	45 491	44.8	46.2	43.9
7	B	12	Wan Chai	灣仔區	166 695	67 075	99 620	673	16 917	9 907	53 860	50 632	35 379	46.0	48.0	44.9
8	C	13	Eastern	東區	529 603	233 542	296 061	789	51 517	36 336	143 911	174 132	123 707	49.0	49.2	48.8
9	D	14	Southern	南區	263 278	116 239	147 039	791	27 255	18 842	72 397	87 976	56 808	48.1	48.6	47.7
10	E	27	Yau Tsim Mong	油尖旺區	310 647	138 873	171 774	808	36 181	23 224	100 877	94 797	55 568	44.0	45.1	43.2
11	F	23	Sham Shui Po	深水埗區	431 090	197 878	233 212	848	49 337	35 476	123 081	135 261	87 935	46.2	46.0	46.3
12	G	24	Kowloon City	九龍城區	410 634	179 518	231 116	777	50 327	31 113	121 174	125 562	82 458	45.4	46.1	45.0
13	H	25	Wong Tai Sin	黃大仙區	406 802	187 531	219 271	855	36 271	33 463	103 363	140 294	93 411	50.1	49.6	50.5
14	J	26	Kwun Tong	觀塘區	673 166	312 303	360 863	865	70 063	60 321	174 338	220 703	147 741	48.0	47.5	48.5
15	S	51	Kwai Tsing	葵青區	495 798	232 213	263 585	881	49 421	42 474	132 936	161 501	109 466	48.0	47.5	48.5
16	K	52	Tsuen Wan	荃灣區	320 094	146 412	173 682	843	37 873	24 515	95 270	104 390	58 046	45.4	45.7	45.3
17	L	53	Tuen Mun	屯門區	506 879	240 321	266 558	902	54 136	38 535	152 609	163 644	97 955	46.1	45.6	46.5
18	M	54	Yuen Long	元朗區	668 080	314 056	354 024	887	77 199	58 988	211 376	220 151	100 366	43.7	43.4	43.8
19	N	55	North	北區	309 631	145 343	164 288	885	33 499	27 654	88 314	104 607	55 557	46.3	46.4	46.2
20	P	56	Tai Po	大埔區	316 470	146 038	170 432	857	34 377	23 198	97 583	102 810	58 502	45.7	45.7	45.6
21	R	57	Sha Tin	沙田區	692 806	317 927	374 879	848	78 886	53 696	201 135	220 604	138 485	46.2	46.0	46.3
22	Q	58	Sai Kung	西貢區	489 037	223 094	265 943	839	56 329	39 744	150 700	164 991	77 273	44.7	44.8	44.7
23	T	59	Islands	離島區	185 282	84 985	100 297	847	23 767	16 808	58 660	58 751	27 296	42.7	43.2	42.4
24	**Z**	**99**	**Land total**	**陸上總計**	**7 411 945**	**3 381 630**	**4 030 315**	**839**	**807 737**	**591 073**	**2 159 351**	**2 402 340**	**1 451 444**	**46.3**	**46.4**	**46.3**
25																
26	註解：	(1) - 數字包括報稱有多於一個種族的人士。														
27	Notes:	(1) - Figures include persons who reported more than one ethnicity.														
28		(2) - 數字不包括失去語言能力的人士。														
29		(2) - Figures exclude mute persons.														
30		(3) - 同區就讀的人士包括（a）在香港島或九龍居住並在同一區議會分區內上課的人士；（b）在新市鎮居住並在同一新市鎮上課的人士，及（c）在新界其他地方居住並在同一區議會分區非新市鎮地區內上課的人士。														
31		(3) - Persons who study in the same district include: (a) those who live on Hong Kong Island or in Kowloon and study in the same District Council district; (b) those who live in a new town and study in the same new town; and (c) those who live in other areas in the New														
32		(4) - 勞動人口佔所有 15 歲及以上人口的百分比。														
33		(4) - The proportion of labour force in the total population aged 15 and over.														
34		(5) - 2021年人口普查所採用非從事經濟活動人口的經濟活動身分分類（即「料理家務者」、「學生」、「退休人士」、「無酬照顧者」及「其他」），跟2011年人口普查及2016年中期人口統計所採用的分類（即「料理家務者」、「學生」、「退休人士」														
35		(5) - The classification of economic activity status for economically inactive population adopted in the 2021 Population Census (i.e. "Home-makers", "Students", "Retired persons", "Unpaid carers" and "Others") slightly differs from that adopted in the 2011 Population C														

Fig 4. Population distribution by district in Hong Kong.

3. Instructions and Assessments

3.1 Instructions

Each individual student is required the accomplish the following 4 tasks:

Task 1 – Data Cleaning

The students will need to clean the data before analyzing them. For example, for all files of indoor and outdoor sports facilities, certain fields with Chinese characters (see Fig. 2) and the columns

where the individual thinks are irrelevant to the analysis part can be removed. Since these datasets contain information about the ***longitude*** and ***latitude*** of the health facilities, the students need to convert such information into a geometry column during the implementation of the database tables. You are also recommended to organize all the files in an identical way (i.e., each type of facility would finally have the same columns) so that they can be handled efficiently.

Task 2 – Database Design and Implementation

Once the data have been cleaned and prepared, each student is required to design a relational dataset with AT LEAST the following three tables:

– A table named ***Facilities*** that records the geographic locations as well as other information about health facilities in Hong Kong.

– A table named ***HK_Admin*** that records the name and the geometry (i.e., polygon) of the administrative districts in Hong Kong. This table can be created from the shapefile of the Hong Kong map.

– A table named ***census*** that records the number and percentage of population in each administrative district. This table should NOT have a geometry column.

Note that the column that describes the name of the administrative districts in ***HK_Admin*** and ***census*** should match with each other, so that they can be joined later when queries are performed (i.e., if you call a district 'Kowloon' in the ***HK_Admin*** table, it should have the same name in the ***census*** table).

You need to draw a schema diagram for your database design, explaining the key concepts such as primary key, relation name, attributes, and foreign key constraints (if any). Please use the university database we have learnt in the class as an example:

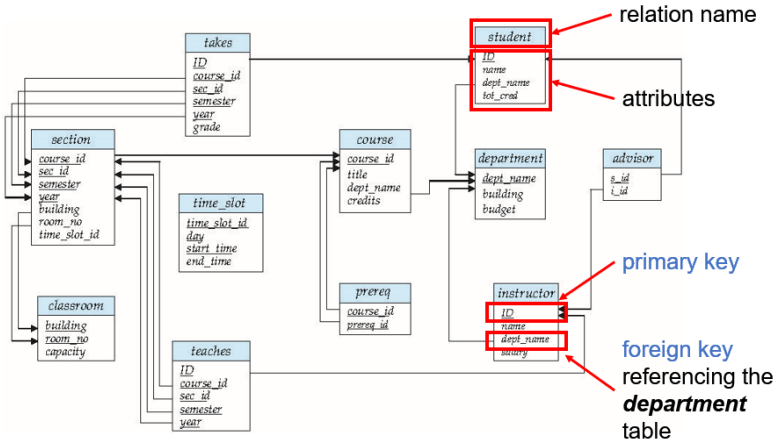


Fig 5. Schema Diagram of the University Database.

Students will then implement the database design using PostgreSQL and PostGIS. The database system created in this step will be used to support spatial and non-spatial queries in the following task.

Task 3 – Perform Spatial and Non-Spatial Queries

In this step, students will need to perform spatial and non-spatial queries to answer a series of

questions about the geographic coverage of health facilities in Hong Kong as well as their relationships with the underlying population distribution. In particular, students need to perform queries to answer the following questions:

– Given a geographic location (e.g., a lat/lng pair), find the nearest health facility and its relevant information (e.g., facility type, name, district).

– Given a geographic location, how many types of sports and outdoor facilities are within a reasonable walking distance? (Students can assume a reasonable walking distance on their own).

– For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the ***number*** of such facilities.

– For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the ***per capita*** of facility count.

3.2 Assessments

Each student needs to DELIVER:

1) A written report (at least two pages) that describes the overall research design and workflow, including data cleaning and preprocessing, the schema design of the relational database, the implementation of the database, and the queries. [40% of the total marks].

2) A folder that includes all the SQL expressions that are used for database implementation and the subsequent queries. The queries should correspond to the four questions mentioned in Task 3. [20% of the total marks].

3) Upon the completion of the above queries, each student should also provide TWO additional questions related to the theme of this topic and explain how they can be answered through SQL queries. Similarly, each student should turn in a folder that includes all the SQL expressions that are used to answer those questions, along with brief instructions on how to use them [15% of the total marks].

4) An oral presentation (15 min) for each group during the class that demonstrates the research design, work flow, and the key findings of the project. A Q&A session will follow the presentation and the audience will be allowed to ask questions. [20% of total marks for presentation, and 5% of total marks for Q&A performance].

4. Important Notes

– Sept 15, 2026: We will release the information of final project using the lecture session (Tuesday)

– Nov 17, 2026 & Nov 24, 2026: We will use lectures in these two weeks for project presentations. Details on the form of presentation will be released later in the semester.

– December 1, 2026 (before 11:59 pm): Submission due for the written report and SQL code.